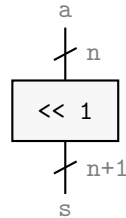
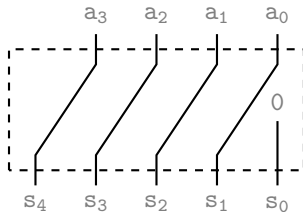


shifters

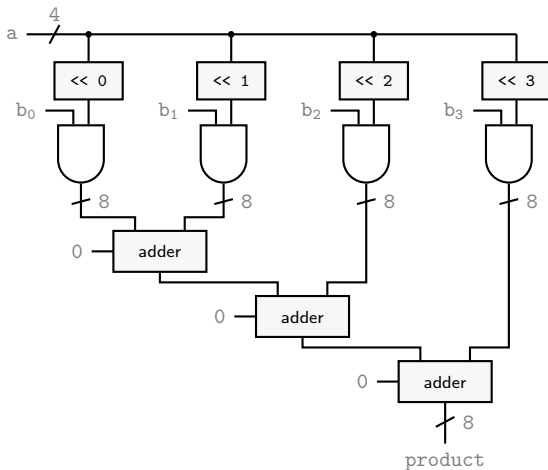
A **shifter** moves every bit one place to the left. A fixed shift needs no gates at all, only wiring.



a shift by k places is the same wiring, moved over k places

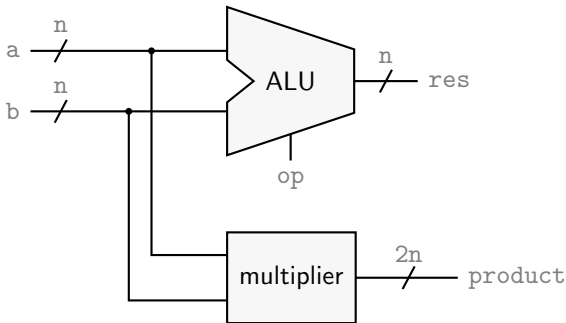
multipliers

In hardware, each row is a passed through a shifter and ANDed with one bit of b . A chain of adders sums the rows.



every cin is 0; the sum always fits in $2n$ bits, so no flags

The multiplier usually sits **beside** the ALU rather than inside it: it is much larger, much slower, and its result is $2n$ bits wide.



a shifter is cheap, so it often becomes one more ALU operation